SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1515

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I ***KESHAVARAM L/192511253*** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student

Assessment Tasks

Scenario-Based Questions

Q1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Answer:

Use a multi-region cloud architecture with object storage for video files, a content delivery network (CDN) for edge caching, and scalable application servers for user requests. Store different video resolutions and use adaptive bitrate streaming.

A load balancer distributes requests across healthy servers. Auto-scaling adds instances when traffic or CPU usage increases and removes them when demand falls. Health checks and multiple availability zones reduce service interruption.

For cost optimization, use CDN caching to reduce origin traffic, object-storage lifecycle policies, reserved capacity for predictable baseline demand, and spot/preemptible instances for suitable background jobs. Monitor bandwidth, cache-hit ratio, latency, and cost to adjust capacity.

Q2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Answer:

Package each service as a versioned container image and store it in a trusted registry. Use Kubernetes or another orchestrator with consistent deployment templates, infrastructure-as-code, and automated CI/CD pipelines across providers.

Service discovery can use Kubernetes Services and DNS, while ingress controllers or service meshes can manage routing and secure service-to-service communication. Horizontal pod autoscaling can scale services based on CPU, memory, or custom metrics.

Benefits include portability, reduced dependence on one provider, and improved resilience. Risks include higher operational complexity, different provider services, data-transfer costs, inconsistent security policies, and difficult cross-cloud networking. Use common policies, centralized monitoring, secrets management, and regular failover tests.

Q3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how RTO and RPO are achieved. Provide steps to ensure business continuity during failures.

Answer:

Enable object versioning and soft-delete or retention protection to recover accidentally deleted files. Schedule automated, encrypted backups and replicate critical data to a separate region or account. Keep an isolated or immutable backup copy to protect against ransomware or compromised credentials.

RTO (Recovery Time Objective) is the maximum acceptable time to restore service; it is reduced by automated recovery, tested runbooks, standby infrastructure, and infrastructure-as-code. RPO (Recovery Point Objective) is the maximum acceptable amount of data loss measured in time; it is reduced by frequent backups and continuous or near-real-time replication.

For continuity, detect the incident, stop further deletion, identify a clean recovery point, restore data and services, validate integrity, redirect users if needed, and communicate status. Test recovery regularly and document responsibilities.

Q4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Answer:

Keep regulated applications and data that must remain local in the on-premises datacenter. Place suitable public-facing or less-sensitive workloads in the cloud. Connect the environments through a private dedicated link or a site-to-site VPN, with firewalls, segmented networks, and restricted routes.

Use a centralized identity provider where possible, federated authentication, role-based access control, least privilege, and multi-factor authentication for administrators. Encrypt communication with TLS or IPsec and encrypt stored data using managed keys with controlled rotation.

Challenges include different security tools, policy consistency, latency, network failures, identity synchronization, monitoring, and cost. Address these through centralized logging, configuration management, regular audits, tested connectivity failover, and clear data-classification rules.

Q5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Answer:

Deploy application instances across multiple availability zones and at least two cloud regions. Use global DNS or a traffic manager to route users to healthy regions. Keep infrastructure configuration consistent through infrastructure-as-code and deploy updates gradually.

Replicate data between regions using synchronous replication when low data loss is essential and latency permits, or asynchronous replication for longer distances. Choose a clear primary/secondary or multi-writer design, define conflict handling, and set RPO and RTO targets. Health checks should trigger automated or controlled failover, with a tested failback plan.

Monitor availability, latency, error rates, CPU, memory, database replication lag, and storage health. Use dashboards, logs, distributed tracing, and alerts with escalation rules. Regularly simulate regional failures to confirm that recovery works.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented